Updated April 17, 2026

Homework problems for AMAT 300 (Intro to Proofs), Spring 2026. Over the course of the semester I'll add problems to this list, with each problem's due date specified. Each problem is worth 2 points.

Solutions will be gradually added (and may be hastily written without proofreading).

Problem 1 (due Weds 2/4): Let $A = \{20a - 26b \mid a, b \in \mathbb{Z}\}$ and $B = \{2c \mid c \in \mathbb{Z}\}$. Prove that $A = B$. (Do it directly, don't use any number theory tricks you might happen to know.)

Solution: ($\subseteq$): Let $x \in A$, so $x = 20a - 26b$ for some $a, b \in \mathbb{Z}$. Set $c = 10a - 13b$, so $c \in \mathbb{Z}$. Now $x = 2c$, so $x \in B$.
($\supseteq$): Let $x \in B$, so $x = 2c$ for some $c \in \mathbb{Z}$. Set $a = 4c$ and $b = 3c$, so $a, b \in \mathbb{Z}$. Then $x = 2c = 80c - 78c = 20(4c) - 26(3c) = 20a - 26b$, so $x \in A$. $\square$

Problem 2 (due Weds 2/4): Prove that the cardinality of $\{1, \{1\}, \{1, \{1\}\}, \{1, 1\}, \{\{1\}, 1\}\}$ is three. (You need to argue that it has exactly three elements, no more, no less.)

Solution: Since $\{\{1\}, 1\} = \{1, \{1\}\}$ and $\{1, 1\} = \{1\}$, we know that our set equals $\{1, \{1\}, \{1, \{1\}\}\}$, hence has at most 3 elements. Also, $1 \neq \{1\}$ since the latter is a set (with one element) and the former is not, which implies $\{1, \{1\}\}$ has two elements, and we conclude that these three elements are distinct. Thus the original set has exactly 3 elements. $\square$

Problem 3 (due Weds 2/4): Let A and B be non-empty finite sets such that $|\mathcal{P}(A \times B)| = |\mathcal{P}(A) \times \mathcal{P}(B)|$. Prove that A and B each have exactly two elements. (Careful not to *assume* they each have exactly two elements, you have to *prove* that they do.)

Solution: Set $a = |\mathcal{P}(A)|$ and $b = |\mathcal{P}(B)|$. Since $|\mathcal{P}(X)| = 2^{|X|}$ and $|X \times Y| = |X||Y|$ for any finite sets X and Y, we compute that $2^{ab} = 2^a 2^b$, hence $2^{ab} = 2^{a+b}$. Taking $\log_2$ we get $ab = a + b$. Since $a, b \in \mathbb{N}$, dividing by a yields $b = 1 + \frac{b}{a}$ and dividing by b yields $a = \frac{a}{b} + 1$, so $\frac{a}{b}$ and $\frac{b}{a}$ are both integers, which implies $a = b$. Now we solve $a^2 = 2a$ to get $a = b = 2$. $\square$

Problem 4 (due Weds 2/11): Let A and B be sets such that $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$ and suppose there exists $b_0 \in B \setminus A$. Prove that $A \subseteq B$.

Solution: Let $a \in A$. Then $\{a, b_0\}$ is a subset of $A \cup B$, hence an element of $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$. Since $b_0 \notin A$ we know $\{a, b_0\}$ is not in $\mathcal{P}(A)$, so it must be in $\mathcal{P}(B)$, i.e., $\{a, b_0\} \subseteq B$. We conclude that $a \in B$ as desired. $\square$

Problem 5 (due Weds 2/11): Let $\Lambda = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$ be the closed disk of radius 1 centered at the origin. For each $\alpha \in \Lambda$, say $\alpha = (x_0, y_0)$, let $X_\alpha = \{(x, y) \in \mathbb{R}^2 \mid (x-x_0)^2+(y-y_0)^2 \leq 1\}$ be the closed disk of radius 1 centered at α. Prove (rigorously) that $\bigcup\limits_{\alpha\in\Lambda} X_\alpha = D$, where $D := \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 4\}$ is the closed disk of radius 2 centered at the origin. [Note: I had a typo initially, writing $= 1$ instead of ≤ 1 in the definition of X_α, but the description "closed disk of radius 1 centered at α" made it clear what I meant and it seems no one got too confused.]

Solution: ($\subseteq$): Let $(x, y) \in \bigcup\limits_{\alpha\in\Lambda} X_\alpha$, so $(x, y) \in X_\alpha$ for some $\alpha \in \Lambda$, say $\alpha = (x_0, y_0)$. Since $\alpha \in \Lambda$ we have $d((0, 0), \alpha) \leq 1$, and since $(x, y) \in X_\alpha$ we have $d(\alpha,(x, y)) \leq 1$ (here d is the usual distance in $\mathbb{R}^2$). Now $d((0, 0),(x, y)) \leq d((0, 0), \alpha) + d(\alpha,(x, y))$ (by the triangle inequality, which is really just that the distance between two points is the shortest length of a path between them), and this is at most 2, so $(x, y) \in D$.
($\supseteq$): Let $(x, y) \in D$, so $d((0, 0),(x, y)) \leq 2$. If $(x, y) \leq 1$ then $(x, y) \in \Lambda$ and $(x, y) \in X_{(x,y)}$ so we are done. Now assume $1 < (x, y) \leq 2$. Let $\alpha = (x, y)/d((x, y),(0, 0))$, so $d(\alpha,(0, 0)) = 1$, which shows $\alpha \in \Lambda$. Moreover, $d((x, y), \alpha) = d((x, y),(0, 0)) - d(\alpha,(0, 0)) \leq 2 - 1 = 1$, so $(x, y) \in X_\alpha$. This shows $(x, y) \in \bigcup\limits_{\alpha\in\Lambda} X_\alpha$, and we are done. □

Problem 6 (due Weds 2/11): With the same setup as the previous problem, prove (rigorously) that $\bigcap\limits_{\alpha\in\Lambda} X_\alpha = \{(0, 0)\}$.

Solution: ($\subseteq$): Let $(x, y) \in \bigcap\limits_{\alpha\in\Lambda} X_\alpha$, so $(x, y) \in X_\alpha$ for all $\alpha \in \Lambda$. In particular (x, y) is in $X_{(-1,0)}$ and $X_{(1,0)}$, so $(x + 1)^2 + y^2 \leq 1$ and $(x - 1)^2 + y^2 \leq 1$. Expanding these and adding them we get $2x^2 + 2 + 2y^2 \leq 2$, hence $x^2 + y^2 \leq 0$, which means $x = y = 0$.
($\supseteq$): We must show that $(0, 0) \in X_\alpha$ for all $\alpha \in \Lambda$. The criterion for α to be in Λ is $d((0, 0), \alpha) \leq 1$, so this is immediate. □

Problem 7 (due Weds 2/18): Construct truth tables for $\sim(P \Leftrightarrow Q)$ and $(P\vee Q)\wedge(\sim(P\wedge Q))$ (showing the intermediate steps), and compare them. (Which logical operation do these represent?)

Solution: Hard to type out truth tables, you get F-T-T-F in both cases, so they're the same, and they represent "exclusive or". □

Problem 8 (due Weds 2/18): Prove that $n \in \mathbb{Z}$ is a multiple of 24 only if it is a multiple of 4 and 6. (Careful, this is not an "iff".)

Solution: Suppose $n \in \mathbb{Z}$ is a multiple of 24, say $n = 24m$ for some $m \in \mathbb{Z}$. Then $n = 4(6m)$ and $n = 6(4m)$, and since $6m, 4m \in \mathbb{Z}$ we conclude that n is a multiple of 4 and 6. □

Problem 9 (due Weds 2/18): Let X be a set. Prove that $|\mathcal{P}(X)| \leq 31$ if and only if $|X| \leq 4$.

Solution: ($\Rightarrow$): Suppose $|\mathcal{P}(X)| \leq 31$, so $2^{|X|} \leq 31$. Since $2^{|X|}$ is 2 to the power of a whole number, in fact $2^{|X|} \leq 16$. Thus $|X| \leq 4$.
($\Leftarrow$): Suppose $|X| \leq 4$. Then $2^{|X|} \leq 16$, so $|\mathcal{P}(X)| \leq 16$, and hence $|\mathcal{P}(X)| \leq 31$. $\square$

(Reminder: There's an exam in class on Weds 2/25.)
Problem 10 (due Weds 2/25): Let P and Q be statements. Prove that $\sim (P \Rightarrow (\sim Q))$ is logically equivalent to $(P \Leftrightarrow Q) \wedge (P \vee Q)$.

Solution: compute their truth tables, they both have T-F-F-F. $\square$

Problem 11 (due Weds 2/25): Let P be the statement, "every set with cardinality 2026 is a subset of a set with cardinality 2027." Convert P into a purely logical statement (using $\forall$, $\exists$, and so forth), and compute its negation. Then prove that P is true.

Solution: $\forall X$, $(|X| = 2026 \Rightarrow \exists Y$ s.t. $(|Y| = 2027 \wedge X \subseteq Y))$. The negation is: $\exists X$ s.t. $(|X| = 2026 \wedge \forall Y, (|Y| \neq 2027 \vee X \not\subseteq Y))$. Prove that P is true: Let X be a set with $|X| = 2026$. Since $\mathbb{N}$ is infinite, $\mathbb{N} \not\subseteq X$, so there exists $n \in \mathbb{N} \setminus X$. Set $Y = X \cup \{n\}$. Now $|Y| = 2027$ and $X \subseteq Y$. $\square$

Problem 12 (due Weds 2/25): Let P be the statement, "every natural number divisible by its own square is divisible by at least two natural numbers." Convert P into a purely logical statement, and prove that P is false.

Solution: $\forall n \in \mathbb{N}$, $(n^2|n \Rightarrow \exists a, b \in \mathbb{N}$ s.t. $a \neq b \wedge a|n \wedge b|n)$. Proof it's false: Set $n = 1$, so $1^2|1$. Let $a, b \in \mathbb{N}$. Then $a|1$ and $b|1$ implies $a = b = 1$, so $a \neq b \wedge a|1 \wedge b|1$ is false. $\square$

Nothing due Weds 3/4

Problem 13 (due Weds 3/11): Let $x \in \mathbb{Z}$. Prove (rigorously) that $2x^2 - 3x + 6$ is even if and only if x is even.

Solution: ($\Rightarrow$): Suppose x is odd, say $x = 2n + 1$ for some $n \in \mathbb{Z}$. Then $2x^2 - 3x + 6 = 2(2n+1)^2 - 3(2n+1) + 6 = 8n^2 + 2n + 5 = 2(4n^2 + n + 2) + 1$, and $4n^2 + n + 2 \in \mathbb{Z}$, so $2x^2 - 3x + 6$ is odd.
($\Leftarrow$): Suppose x is even, say $x = 2n$ for some $n \in \mathbb{Z}$. Then $2x^2 - 3x + 6 = 2(2n)^2 - 3(2n) + 6 = 8n^2 - 6n + 6 = 2(4n^2 - 3n + 3)$, and $4n^2 - 3n + 3 \in \mathbb{Z}$, so $2x^2 - 3x + 6$ is even. $\square$

Problem 14 (due Weds 3/11): Let $x, y \in \mathbb{R}$. Consider the conditional statement, "if $x < y$ then for all $z \in \mathbb{R}$ either $z > x$ or $z < y$." (Part (a)): Prove this directly. (Part (b)): Prove this using the contrapositive. (Which one do you think is more elegant? They seem about the same to me.)

Solution: (Direct): Suppose $x < y$. Let $z \in \mathbb{R}$. If $z \le x$ then $z < y$, so we conclude that $z > x$ or $z < y$. $\square$

(Contrapositive): Suppose $z \le x$ and $z \ge y$. Then $x \ge z \ge y$ so $x \ge y$. $\square$(I guess I like this one better?)

Problem 15 (due Weds 3/11): Let $a \in \mathbb{N}$. Prove that $a^3 - a - 1$ is not divisible by 3.

Solution: Suppose $a^3 - a - 1$ is divisible by 3, say $a^3 - a - 1 = 3n$ for some $n \in \mathbb{Z}$. Since $a^3 - a = (a-1)a(a+1)$ we get $(a-1)a(a+1) = 3n+1$. Since $a-1, a, a+1$ form a sequence of three successive integers, exactly one of them is divisible by 3, so $3m = 3n + 1$ for some $m \in \mathbb{Z}$. Now $1 = 3(m - n)$, so 1 is divisible by 3, a contradiction. $\square$

Problem 16 (due Weds 3/25):
Prove that there do not exist $a, b \in \mathbb{Z}$ such that $2026a + 3039b = 1014$.

Solution: Suppose there do. Then $1013 + 1 = 1013(2a + 3b)$, so $1 = 1013(2a + 3b - 1)$. But this implies 1 is divisible by 1013, a contradiction. $\square$

Problem 17 (due Weds 3/25):
Prove that there exists $n \in \mathbb{Z}$ such that $-2026 < \frac{\cos(n+7)}{31e^n} \cdot \frac{n^{1/3}}{2\pi} < \frac{1}{2027}$.

Solution: Set $n = 0$. Then $\frac{\cos(n+7)}{31e^n} \cdot \frac{n^{1/3}}{2\pi} = 0$, and $-2026 < 0 < \frac{1}{2027}$. $\square$

Problem 18 (due Weds 3/25): Let $S \subseteq \mathbb{R}$. Call $s \in S$ a *maximum* element of S if $s' \le s$ for all $s' \in S$. Prove that if S has a maximum element then it has a unique maximum element.

Solution: Let s_1 and s_2 be maximal elements of S. Since s_1 is maximal and $s_2 \in S$ we know $s_2 \le s_1$, and since s_2 is maximal and $s_1 \in S$ we know $s_1 \le s_2$. We conclude $s_1 = s_2$. $\square$

(Reminder: There's an exam in class on Weds 4/1.)
Problem 19 (due Weds 4/1): Use induction to prove that $n^2 \ge n + 2026$ for all $n \ge 46$.

Solution: As a base case use $n = 46$, and note that $(46)^2 = 2116 \ge 2072 = 46 + 2026$. Now suppose $n \ge 47$ and assume for induction that $(n-1)^2 \ge (n-1) + 2026$. This implies $n^2 \ge 3n - 2 + 2026$, and since $2n - 2 \ge 0$ this implies $n^2 \ge n + 2026$. $\square$

Problem 20 (due Weds 4/1): Use (strong) induction to prove that $3|(n^4 + n^3 - n^2 - n)$ for all $n \in \mathbb{N}$. (I think three bases cases works best.)

Solution: Our base cases are $n = 1, 2, 3$. We check that in these cases $n^4 + n^3 - n^2 - n$ equals 0, 18, and 96, which are all divisible by 3. Now suppose $n \geq 4$ and assume for induction that $3|((n-3)^4 + (n-3)^3 - (n-3)^2 - (n-3))$. Expanding this out (skipping scratch work) we get that $3|(n^4 - 11n^3 + 44n^2 - 76n + 48)$, say $n^4 - 11n^3 + 44n^2 - 76n + 48 = 3k$ for some $k \in \mathbb{Z}$. Now $n^4 + n^3 - n^2 - n = 3k + 12n^3 - 45n^2 + 75n - 48 = 3(k + 4n^3 - 15n^2 + 25n - 16)$, and $k + 4n^3 - 15n^2 + 25n - 16 \in \mathbb{Z}$, so $3|(n^4 + n^3 - n^2 - n)$. $\square$

Problem 21 (due Weds 4/1): Prove that for every $n \geq 20$, one can pay exactly n cents in postage using only 5-cent and 6-cent stamps.

Solution: Use five base cases, $n = 20, 21, 22, 23, 24$. We can write each of these in the form $n = 5k + 6\ell$ for $k, \ell \in \mathbb{N} \cup \{0\}$ using $(k, \ell) = (4, 0), (3, 1), (2, 2), (1, 3), (0, 4)$ respectively. Now suppose $n \geq 25$ and assume for induction that $n - 5 = 5k' + 6\ell'$ for some $k', \ell' \in \mathbb{N} \cup \{0\}$. Set $k = k' + 1$ and $\ell = \ell'$. Now $5k + 6\ell = 5 + 5k' + 6\ell' = 5 + (n - 5) = n$, and we are done. $\square$

Nothing due Weds 4/8 (and no class that day).

Problem 22 (due Weds 4/15): Use induction to prove that $17|(35^n + 16)$ for all $n \in \mathbb{N}$. [Hint: Don't simplify products, it'll just make things harder to detect divisibility later.]

Solution: The base case is that $17|(35 + 16) = 51 = 17 \cdot 3$, which is true. Now suppose $n \geq 2$, and assume for induction that $17|(35^{n-1} + 16)$, say $35^{n-1} + 16 = 17k$ for $k \in \mathbb{Z}$. Multiplying both sides by 35 we get $35^n + 35 \cdot 16 = 35 \cdot 17k$, so $35^n + 16 = 35 \cdot 17k - 34 \cdot 16 = 17(35k - 2 \cdot 16)$. Since $35k - 2 \cdot 16 \in \mathbb{Z}$, we conclude $17|(35^n + 16)$. $\square$

Problem 23 (due Weds 4/15): Let R be the relation on $\mathbb{Z}$ defined by: xRy whenever $x^2 + y^2 \leq 9$. Prove that R is symmetric, but neither reflexive nor transitive.

Solution: It is symmetric for trivial reasons. To see it is not reflexive, note that $3^2 + 3^2 = 18 \not\leq 9$, so $3R3$ is false. Finally, to see it is not transitive, note that $3R0$ and $0R3$ since $3^2 + 0^2 = 0^2 + 3^2 = 9 \leq 9$, but we saw $3R3$ is false. $\square$

Problem 24 (due Weds 4/15): Let R be a reflexive symmetric relation on a set A. Let Q be the relation defined via: aQb whenever there exist $a = a_0, a_1, \ldots, a_n = b$ in A such that $a_i R a_{i+1}$ for all $0 \leq i < n$. Prove that Q is an equivalence relation. [Careful to actually prove Q is still reflexive and symmetric (and now transitive).] [This one might be a little harder than usual!]

Solution: Reflexive: Let $a \in A$. Since R is reflexive, aRa. Now using $a = a_0, a_1 = a$ since aRa we get aQa.
Symmetric: Suppose aQb, so there exist $a = a_0, a_1, \ldots, a_n = b$ in A such that $a_i R a_{i+1}$ for all $0 \leq i < n$. For each i set $a'_i = a_{n-i}$, so $b = a'_0, a'_1, \ldots, a'_n = a$. Since R is symmetric we know $a_{n-i} R a_{n-i-1}$ for all $0 \leq i < n$, so $a'_i R a'_{i+1}$ for all $0 \leq i < n$, which implies bQa.

Transitive: Suppose aQb and bQc, say there exist $a = a_0, a_1, \ldots, a_n = b$ and $b = b_0, b_1, \ldots, b_m = c$ in A such that $a_i R a_{i+1}$ for all $0 \leq i < n$ and $b_i R b_{i+1}$ for all $0 \leq i < m$. Since $a_n = b_0$, the sequence $a = a_0, a_1, \ldots, a_{n-1}, b_0, b_1, \ldots, b_m = c$ satisfies the desired property to reveal that aQc. $\square$

Problem 25 (due Weds 4/22): Let $f\colon \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ be the function defined by $f(m,n) = 20m + 26n$. Prove that f is neither injective nor surjective.

Problem 26 (due Weds 4/22): Let A be a subset of $\{1, \ldots, 19\}$ with exactly 7 elements. Prove that there exist subsets $X \neq Y$ of A such that the sum of the elements of X equals the sum of the elements of Y.

Problem 27 (due Weds 4/22): Let $f\colon X \to Y$ be a surjective function and let Π be a partition of Y. For each $B \in \Pi$ let $\widetilde{B} := \{x \in X \mid f(x) \in B\}$. Prove that $\Psi := \{\widetilde{B} \mid B \in \Pi\}$ is a partition of X.

(Reminder: There's an exam in class on Weds 4/29.)
Problem 28 (due Weds 4/29):

Problem 29 (due Weds 4/29):

Problem 30 (due Weds 4/29):

End of homework.